

Flying high The launch of Vikram-1, India's first privately developed orbital-class rocket, built by Hyderabad-based firm Skyroot Aerospace

PHOTO BY **ANI**

earth's rotation and, therefore, appear stationary from the ground.

Abyom is another example. A reignitable, throttleable engine didn't exist in India, and it couldn't be reverse-engineered. "Rocket propulsion is fundamentally a deep-tech problem. You cannot simply take an existing rocket engine apart and reverse-engineer it like you might with a conventional automobile engine," says Abedin. "Every element—from combustion and injection to cooling, materials and controls—has to work together under extreme conditions. We have had to build a significant part of this capability ourselves, through continuous testing, iteration and learning."

The gap was widest where the work was design-intensive. Lenses and sensors, for example, need a specialised supply chain: makers of optical glass, of the fine coatings that go on lenses, of ultra-precise polishing and measurement. Almost none of it exists in India. So EON had to build the know-how itself. "We're no longer just assembling imported components," says Co-founder and COO Punit Badeka. "We're designing and engineering sophisticated electro-optical payloads in India."

How much a company inherited then depended on which part of space it had chosen. The closer their work sat to hardware Isro had already built, the more there was to draw on; the further it moved into new territory, the less.

Digantara, a space domain awareness and surveillance start-up, sits at the far end. "We had to build every bit of infrastructure from scratch," says the Bengaluru-based company's founder and CEO Anirudh Sharma. "Isro had built satellites, rockets and missions, but it had never tracked satellites from space." There was no vendor to call, no scientist to borrow, no expertise anywhere in the country to hire.

Between the two ends sat most of the industry, drawing on what existed and building what did not. The supplier base gave the new companies manufacturing capacity. Isro gave them people, testing infrastructure, engineering experience and credibility. The policy reforms gave them permission to become mission owners. But the technology they now had to own, and the supply chains they needed to make it at scale, had to be built by the companies themselves.

Vikram-1 showed that India's private sector could assemble an orbital vehicle and put it into space. The next test is whether India's spacetech companies can compete globally. BT